When Averaged Field Regularity Fails to Rank Few-Step Generative Paths

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Abstract

Choosing a probability path for few-step generative sampling is often guided by averaged regularity of the velocity field: a path with smaller averaged squared Jacobian norm is expected to discretize better at a fixed budget of function evaluations (NFE). We study this question in commuting Gaussian flows, where endpoint laws and Gaussian 2-Wasserstein (W_2) errors are available in closed form and the regularity integrand is analytically known; sampling, training, and estimation noise are absent. The averaged-regularity ordering of linear versus variance-preserving paths fails to determine the fixed-NFE W_2 ordering: fourteen inversions identified in a registered benchmark reproduce in a frozen clean-code grid across Euler, Heun, and RK4, survive an 80-digit precision audit, and recur in a separately pre-registered non-centered replication family. An exact modal decomposition expresses the mismatch through solver-stage sampling and signed one-step defects transported to the endpoint; an explicit scalar construction proves the non-implication for fixed-grid Euler. This is a controlled limitation study: it does not refute regularity-guided schedule design, and no learned model is evaluated.

1 Introduction

Few-step sampling of continuous-time generative models commits to a probability path, a solver, and a small NFE budget before any error is observed. Lipschitz and stability constants enter classical numerical-error bounds [Hairer et al., 1993], and Chen et al. [2025] propose averaged squared drift Lipschitzness as a schedule criterion; bounding an error, however, is weaker than determining which of two paths errs less at an equal small budget. We isolate that question where the marginal flow is affine, the fixed-step endpoint law follows in closed form, and the error is the closed-form Gaussian W_2 [Gelbrich, 1990]; reported values are checked against an 80-digit reference.

Contributions.

- A controlled closed-form W_2 benchmark on commuting Gaussian flows: fourteen inversions identified in a registered benchmark reproduce in a frozen clean-code grid (14 of 36 comparisons, 80-digit-verified), and a separately pre-registered non-centered replication family shows the same phenomenon (11 of 18).
- An exact eigenmode analysis: the endpoint factor error is a transported sum of signed, solver-stage-dependent local defects, which an unsigned time average of Jacobian norms cannot retain.
- An explicit grid-aware smooth scalar construction proving that, for every $L > 0$ and grid size N, strictly larger averaged squared Jacobian is compatible with zero fixed-grid Euler endpoint error.

Solver dependence of good schedules is not itself novel [Sabour et al., 2024, Karras et al., 2022]; the contribution is the exact, controlled non-implication and its modal explanation, bounding what averaged regularity certifies about fixed-budget *rankings* in the tested systems.

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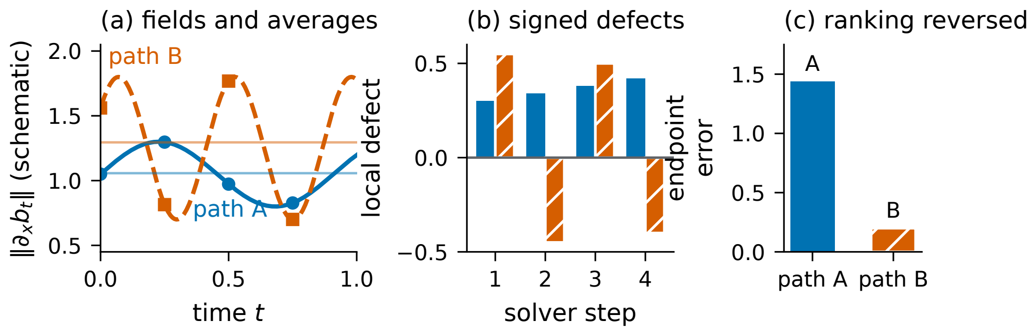

Figure 1: **Conceptual schematic, not experimental data.** (a) Averaging collapses two time-dependent fields to two scalars, while a solver samples only stage locations (dots and squares). (b) Local defects are signed: same-sign defects accumulate, alternating defects cancel. (c) Transported to the endpoint, the error ranking can oppose the averaged ranking. This illustrates the accounting of Section 3, not the measured mechanism of any specific Gaussian inversion.

2 Gaussian probability flows and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent and let a scalar schedule $(\alpha(t), \sigma(t))$, with $\alpha(0) = \sigma(1) = 1$ and $\alpha(1) = \sigma(0) = 0$, define the interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023]. The marginal law is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q(t) = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$, and the marginal velocity field is affine,

$$b(t, x) = A(t)\, x + c(t), \qquad A(t) = C_t\, Q(t)^{-1}, \qquad c(t) = \dot{m}_t - A(t)\, m_t, \tag{1}$$

with $C_t = \dot\alpha\alpha\,\Sigma_0 + \dot\sigma\sigma\,\Sigma_1$; centered with $\Sigma_0 = I$ this is $A(t) = \frac{1}{2}Q'Q^{-1}$, $c \equiv 0$. We compare the *linear* path $(1-t,\, t)$ and the trigonometric *variance-preserving* (VP) path $(\cos\frac{\pi t}{2},\, \sin\frac{\pi t}{2})$, the interpolant analogue of VP diffusion schedules (background: Song et al., 2021). Following Chen et al. [2025], the baseline is $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2\, dt$, evaluated by trapezoidal quadrature of the analytically known spectral-norm integrand; by its own definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2, the closed-form distribution-level metric of this schedule-design literature.

A fixed-step solver applied to (1) induces an affine endpoint map, reconstructed from $d{+}1$ deterministic probes; pushing the source moments through it and applying the closed-form Gaussian W_2 [Gelbrich, 1990] gives the endpoint error (float64, verified against an 80-digit reference below). With $\Sigma_0 = I$ and $\Sigma_1 = U\mathrm{diag}(\lambda_i)U^\top$ all matrices commute: each mode obeys $x_i' = a_i(t)x_i + c_i(t)$ with $a_i = q_i'/(2q_i)$, $q_i = \alpha^2 + \sigma^2\lambda_i$, and centered, $W_2^2 = \sum_i(|r_i| - \sqrt{\lambda_i})^2$ with r_i the product of the numerical one-step factors of mode i.

3 Why the ranking need not agree: transported signed defects

The exact one-step transition of mode i over $[t_n, t_{n+1}]$ is $e_{i,n} = \sqrt{q_i(t_{n+1})/q_i(t_n)}$, while a Runge–Kutta step multiplies by a factor $r_{i,n}$ built from stage samples of a_i (one for Euler, two for Heun, four for RK4). For smooth a, the leading coefficients of the local defect (exact minus method) are $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-\frac{1}{12}a'' + \frac{1}{6}a^3)h^3$ for Heun—different derivatives, signs, and stage locations. With S the number of steps, the endpoint factor error obeys the exact telescoping identity

$$\prod_{n<S} r_{i,n} - \prod_{n<S} e_{i,n} = \sum_{n<S} (r_{i,n} - e_{i,n}) \Big(\prod_{j<n} r_{i,j}\Big)\Big(\prod_{j>n} e_{i,j}\Big), \tag{2}$$

with no smallness or positivity assumption. This identifies what $\mathcal{R}$ discards: which times the stages sample, the derivatives of A, the *signs* of local defects, their cancellation, and their transport to the endpoint—so an unsigned average of $\|A\|^2$ need not determine the fixed-NFE ordering.

Table 1: Strongest inversion: low-rank Gaussian, $d = 8$, Euler, NFE 8 (artifact A1).

path	$\mathcal{R}[b]$ (lower = preferred)	Gaussian W_2
linear	$\mathbf{2.9476523251}$	0.8108540111
variance-preserving	4.7295206355	$\mathbf{0.4564779075}$

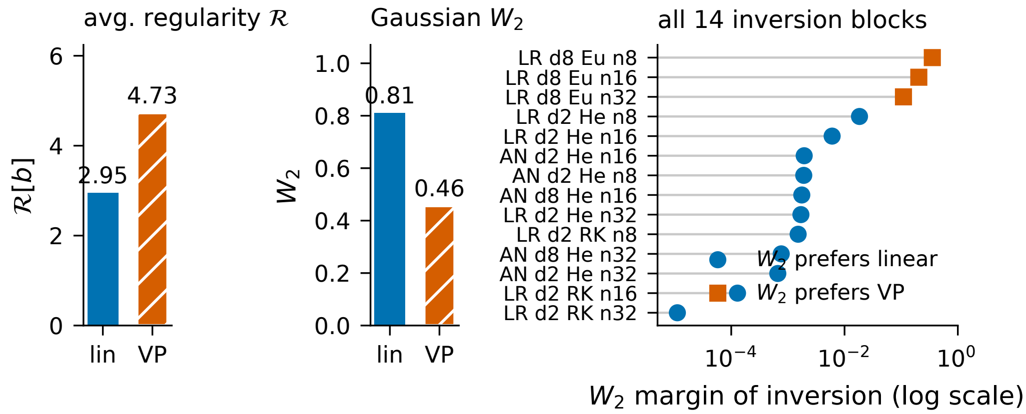

Figure 2: **Left:** the strongest inversion in both quantities; hatching marks VP. **Right:** W_2 margins of all fourteen inversion blocks on a logarithmic axis (which compresses, never exaggerates, magnitudes); marker shape and color give the W_2-preferred path (artifact A1).

4 Controlled benchmark and results

Design and chronology. Source $N(0, I)$; anisotropic (condition number 4) and seeded low-rank ($FF^\top + 0.05I$, rank 2) targets; $d \in \{2, 8\}$; linear and VP paths; Euler, Heun, RK4; NFE $\in \{8, 16, 32\}$ with equal-NFE accounting (steps $=$ NFE$/$stages): 72 float64 CPU configurations in 36 two-path blocks. A *ranking inversion* is a block where the $\mathcal{R}$ and W_2 orderings strictly disagree. The inversions were first identified in a registered benchmark, then the grid above was frozen and rerun from clean code; post-hoc diagnostics are labeled as such.

Fourteen audited inversions. All 72 configurations reproduce the registered run within 10^{-12}; 14 of 36 blocks are inversions (Figure 2; audit artifact A1). In the strongest, low-rank at $d = 8$ with Euler at NFE 8 (Table 1), the baseline prefers linear whose W_2 is 78% larger (margin 0.3543761036). All 72 float64 W_2 values agree with an 80-digit `mpmath` reference to at most 9.71×10^{-10}; the smallest inversion margin, 1.1189×10^{-5}, exceeds this 11,500-fold (artifacts A1–A2).

Solver–path interaction. In the low-rank grid, VP is preferred under Euler and linear under Heun and RK4, at both dimensions and every primary budget (Figure 3); the pattern persists at NFE 64 and 128 and under $\pm 10\%$ target perturbations (artifact A4). Solver-dependent schedule preference is established [Sabour et al., 2024]; here, the tested averaged-regularity scalar does not capture this interaction.

Pre-registered non-centered replication. Before execution we froze one supporting family: source $N(\mu_0, I)$, $\mu_{0,i} = 0.75(-1)^i$; target $N(\mu_1, D)$, $\mu_{1,i} = 1 + 0.25i$, D geometric with condition number 6—so $c(t) \neq 0$ in (1)—with the same paths, solvers, and budgets. Eleven of eighteen blocks are inversions ($\mathcal{R}$, which ignores $c(t)$ by definition, prefers VP with margin 0.7513; W_2 prefers linear), all passing the 80-digit audit (max float64 deviation 2.07×10^{-11}; smallest inverted margin 1.42×10^{-6}; artifacts A5–A7). The family remains Gaussian, diagonal, and commuting: a single supporting replication for the tested systems, not out-of-distribution evidence. No mixture target supports any conclusion here; mixture evidence failed estimator calibration and was excluded.

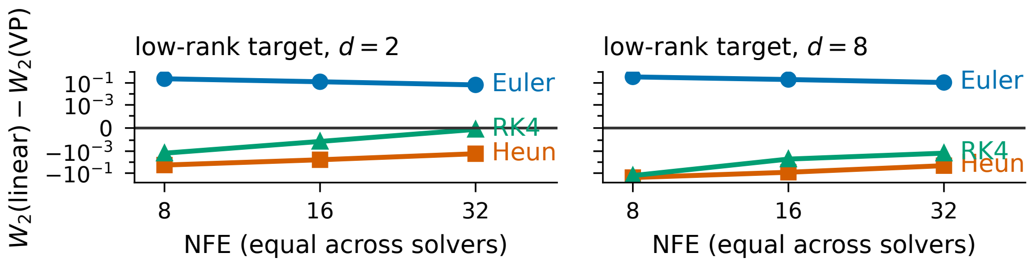

Figure 3: Signed low-rank W_2 margin at equal NFE (artifact A1). Positive values prefer VP; negative values prefer linear; the preferred path flips with the solver. The axis is symmetric-log (linear threshold 10^{-4}) since margins span four orders of magnitude and change sign.

5 A minimal non-implication for fixed-grid Euler

Proposition 1. *Fix $L > 0$ and an integer $N \geq 1$; let $h = 1/N$ and let forward Euler use the left-endpoint grid $t_n = nh$ on $[0, 1]$. Set $\epsilon_N = N\{e^{L/N} - 1\} - L > 0$ and define $f_k(t, x) = a_k(t)x$ with $a_0(t) = L$ and $a_{1,N}(t) = L + \epsilon_N \cos(2\pi N t)$. For $x(0) = 1$: (i) both exact endpoints equal e^L; (ii) $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2 > \int_0^1 a_0^2$; (iii) Euler is* exact *for $a_{1,N}$; (iv) Euler has strictly positive endpoint error for a_0. Hence larger averaged squared Jacobian does not imply larger fixed-grid Euler endpoint error.*

Proof. (i) $\int_0^1 \cos(2\pi N t)\, dt = 0$, so both coefficients integrate to L and $x(1) = e^L$. (ii) Orthogonality gives $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2$, and $\epsilon_N > 0$ since $e^z > 1 + z$ for $z > 0$. (iii) $\cos(2\pi N t_n) = 1$ at every node, so each Euler factor is $1 + h(L + \epsilon_N) = e^{L/N}$; the product is e^L. (iv) $(1 + L/N)^N < e^L$ since $\log(1 + z) < z$. $\square$

The quantifiers matter: $a_{1,N}$ is chosen after the grid, its amplitude vanishes as $N \to \infty$, and shifted grids or multi-stage solvers sample different phases. The construction shows non-implication for fixed-grid left-endpoint Euler only—neither the mechanism behind every Gaussian inversion nor a new numerical-analysis phenomenon; the full proof and edge-case audit are in the supplement.

6 Related work

Flow matching and stochastic interpolants define the Gaussian path family and its fields [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023]; Lipschitz-guided schedule design proposes the criterion we test [Chen et al., 2025]. Fast diffusion solvers exploit exactly integrated linear parts and solver-specific expansions [Lu et al., 2022a,b, Zhang and Chen, 2023], EDM separates schedule from discretization [Karras et al., 2022], and Align Your Steps tunes schedules per solver, model, and dataset [Sabour et al., 2024]: solver-dependent schedule quality is known. Runge–Kutta and backward error analysis supply the local-error machinery [Butcher, 1963, Hairer et al., 1993, 2006]; Gaussian W_2 is due to Gelbrich [1990]. None states the commuting inversion benchmark or the grid-aware non-implication; search absence does not establish novelty.

7 Limitations and conclusion

Our systems are closed-form, commuting, affine Gaussian flows with two paths, three solvers, $d \in \{2, 8\}$, and small budgets, plus one pre-registered non-centered replication family; no learned neural field is evaluated and the proposition is grid-aware. Exponential and exact-linear integrators solve affine fields exactly, so the result concerns generic fixed-stage Runge–Kutta discretization. Our result does not refute Lipschitz-guided schedules: in the tested systems the averaged-regularity *ordering* does not determine the fixed-NFE W_2 ordering, because the tested averaged-regularity scalar does not retain the temporal information carried by solver-stage sampling and transported signed defects. All numerical statements map to the accompanying audit artifacts.

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